

EXP-014R — DOC-C Replication and Golden Set Expansion

Production RAG v1 · evaluation-first · systems frozen before any new data

STATUS — THE REPLICATION COULD NOT BE PERFORMED, AND THE REASON IS THE FINDING

The systems were frozen and hashed, the evaluation infrastructure was built and works, and the harness reproduces EXP-014 exactly. But the expanded golden set the replication depends on **could not be built to a usable standard**, and running a replication on the set that was generated would have produced a confident number with nothing behind it.

[+0.000, +0.250]

bootstrap 95% CI on the EXP-014 delta

lower bound touches zero

p = 0.5

McNemar exact, 2 discordant pairs

12

usable generated questions against a minimum of 100

1

real defect found in the original human-verified set

1. What was delivered

deliverable	status
SYSTEM-A and SYSTEM-B frozen and hashed	done
Golden-set schema, splits, provenance fields	done
validate_golden.py with 16 blocking checks	done
Source-anchored candidate generator	done — and it demonstrates why the approach fails
Development split, verbatim, hashed	done (22 cases)
Replication harness, bootstrap + McNemar	done, verified against EXP-014
Expanded validation/holdout splits (100–150)	NOT delivered
Holdout replication of DOC-C	NOT run

2. Frozen systems

SYSTEM-A-GLOBAL	9afcb5b7c58ebacf...	global BM25 + transformer RRF
SYSTEM-B-DOC-C	304c350940b83733...	DOC-C-SECTION routing -> frozen Stage 2

Both frozen **before** any new evaluation data existed. Recorded metrics are excluded from the hash (an outcome is not a setting) and a test proves that changing `top_documents` *does* change it. The secondary 19/20 router is recorded as explicitly not under test, with its reason: it routes better and retrieves worse.

3. Why the golden set could not be expanded

The brief permits automated generation *with human verification*. **There is no human in this session**, so the honest ceiling on any question I generate is `source_anchored_automatic`, never `human_verified`. That alone would have been

survivable. Quality was not.

pass	yield	representative failure
1. broad patterns	123	"Which HTTP status code corresponds to the ptimized condition ?" → claim 512 . A truncated word, and a status code invented by pairing an unrelated number with a nearby token.
2. + identifier shape, + soundness check	45	"maximum value allowed for resold " — a word fragment as a parameter. rbac_group_id → 100 , where the 100 governed something else.
3. + literal-value filter, limit pattern removed	20	tool_choice → True (wrong). effort given two contradictory answers . max_concurrent_subagents → used .

Each tightening cut yield without eliminating false facts, ending at **12 supported questions** — several still wrong — against a stated minimum of 100. The failure is structural: extracting "X defaults to Y" needs to know which X a sentence is *about*, and regexes bind to the nearest plausible token — right often enough to look fine in aggregate, wrong often enough to poison an answer key.

WHAT WAS DELIBERATELY NOT DONE

No replication was computed on those 12 questions. Reporting "DOC-C replicated on n=12" would have been worse than reporting nothing: **an evaluation set is the one artifact that cannot be checked by running it**, so a wrong key yields a confident number with nothing behind it.

4. The validator works — it blocked the run

20 cases, 13 failures

check	failures
human_verified_required	12
duplicate_question	1

The duplicate it caught is the contradictory `effort` pair. The human-verification gate is what stops machine-generated ground truth from silently becoming gold. Sixteen checks are implemented, including **evidence span hash matches** and **every critical claim appears in its own evidence span**.

5. A real defect in the original golden set

OA-002 — CLAIM ABSENT FROM ITS OWN EVIDENCE

"Which exception does the OpenAI Agents SDK raise when a run exceeds the `max_turns` limit?", critical claim `MaxTurnsExceeded` . The cited span *describes* the exception without naming it; the name occurs elsewhere in the document and the anchor's start boundary excludes it.

A system returning exactly this span scores as having found the evidence. One of 22 spans — up to ~4.5 percentage points, about one case, on every experiment since EXP-000. **It has not been fixed**: correcting it now would silently change every historical number without re-running anything.

6. Harness verification, and the statistics that matter

system	macro recall	full	spans@10	doc R	MRR
SYSTEM-A-GLOBAL	0.775	15/20	17/22	0.925	0.449
SYSTEM-B-DOC-C	0.875	17/20	19/22	0.925	0.474

Reproduces EXP-014 exactly — B rescues AN-006, AN-011, **zero regressions**, net +2. This is the *development* set: a reproduction, not a replication.

THE EXP-014 RESULT IS NOT DISTINGUISHABLE FROM ZERO ON ITS OWN DATA

Macro-recall delta **+0.100**, bootstrap 95% CI [**+0.000, +0.250**] (10,000 paired resamples of questions, seed 20250818). McNemar exact **p = 0.5** on 2 discordant pairs.

Two rescues out of twenty is exactly the evidence you would expect from a real +0.100 effect *and* from a coin landing twice — the data cannot separate them. The bootstrap resamples questions, not spans, because spans within a question are not independent.

7. Two requested categories are not buildable from this corpus

- **version_conflict** — the snapshot contains **zero** superseded versions, so "current vs superseded" cannot be anchored. 52 documents discuss deprecation within a single version, which is a weaker and different thing.
- **routing_heavy / passage_heavy** — properties of how the *systems* behave, not of the corpus. Labelling them requires running the systems first, which is the leakage the split design exists to prevent.

The corpus is also 139 Anthropic / 63 OpenAI, and extraction yield was worse for OpenAI.

8. Promotion decision

DOC-C is not promoted. The frozen production baseline remains SYSTEM-A-GLOBAL. EXP-014's +2/0 stands as recorded, on the development set, with a confidence interval whose lower bound is zero — a promising candidate, never eligible for promotion without replication.

9. What the measurements justify next

1. **The blocker is human question authoring, and it cannot be automated away.** ~100–150 source-anchored questions need a person. Everything needed to consume them now exists: schema, splits, manifests, validator, harness, bootstrap, McNemar, frozen hashed systems.
2. **Change the generator's role** to proposing candidate *evidence spans* for a human to write against. Span extraction was reliable; question and claim synthesis was not.
3. **Decide about OA-002 explicitly** — accept a known ~4.5-point defect, or create `development/v2` and re-run. Do not fix it silently.
4. **Do not run more retrieval experiments against n=20.** Six of the last seven turned on one or two cases, and the interval on such a result includes zero.
5. **If the corpus is re-fetched, capture version chains** — without superseded versions the project cannot test the temporal behaviour it was designed for.